

# StressLess: Towards Generalizable Fragile Food Grasping from Stress-Aware Demonstrations at Scale

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**Abstract**—Handling fragile food requires a robot to select both where to grasp and how far to close the gripper, small errors in either decision can cause irreversible damage. Learning this behavior from human demonstrations is costly because achieving coverage across object categories, poses, sizes, shapes, and materials requires extensive physical data collection. Towards addressing this challenge, we present METHOD, a framework that bootstraps gentle manipulation policies from stress-aware demonstrations generated at scale in simulation. A finite-element surrogate predicts the internal stress induced by gripper closure, and a scripted planner jointly selects the grasp pose and closing width that minimize stress while maintaining sufficient grip to resist gravity. By massive randomization, METHOD generates over 6k diverse demonstrations across 34 object categories without any human tele-operation. We use these data to train a single point-cloud-conditioned diffusion policy and deploy zero-shot in real-world. Real-robot experiments achieve a [XX%] success rate across 11 categories. These results show that physics-based simulated supervision can efficiently provide the broad coverage required for generalizable gentle manipulation.

## I. INTRODUCTION

Humans handle delicate food almost effortlessly, yet the same task remains challenging for robots. Automating this capability is important in food processing, agriculture, and assistive healthcare, where compression can tear, crack, or bruise produce and cause irreversible damage [1], [2], [3]. Gentle grasping requires the robot to jointly select the contact configuration, where and at what orientation the fingers close, and an appropriate closing width. These decisions are coupled: a closing width that is appropriate at one location may cause failure across a thin edge or when the gripper straddles a stem. Moreover, both decisions depend on the item’s pose, size, and shape. These properties vary continuously within a food category and can differ entirely across categories, so success on one item does not necessarily transfer to another.

Existing approaches address this challenge primarily through hardware design and contact sensing. Compliant and pneumatic end-effectors mechanically limit contact pressure [4], while tactile or force-controlled grippers regulate

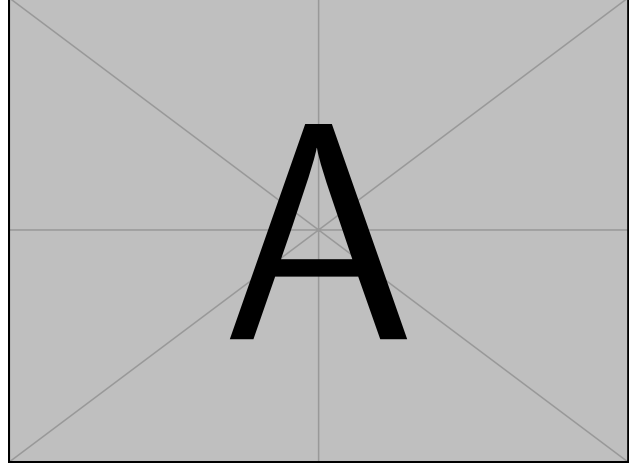


Fig. 1. Teaser.

contact at run time and can handle delicate items effectively [5], [6]. These approaches improve how contact is regulated or observed, but they do not address the source of policy *supervision*: policies are still learned by imitation from human demonstrations recorded on real hardware, one item and one pose at a time. Achieving *broad coverage* of object categories and physical variations therefore remains a human-labour intensive problem.

The importance of such coverage is supported by recent scaling results. Lin *et al.* [7] show that policy performance follows a power law as the number of environments and objects increases, whereas the benefit of additional demonstrations per object saturates quickly. Thus, *diversity*, rather than training data size alone, drives generalisation. The data requirement becomes even more severe for high-precision tasks: the number of demonstrations grows super-exponentially as the tolerance tightens, while the limiting precision itself depends on the quality of the expert [8]. Existing data-generation systems reduce collection costs by adapting a few hundred human demonstrations to new object placements, producing tens of thousands of episodes [9]. However, this adaptation is geometric and preserves the grasps selected by a human; it cannot determine anew how firmly to close on an unseen item.

To address this limitation, we propose METHOD, a framework that autonomously generates gentle demonstrations in simulation and uses them to train a single policy across

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multiple fragile food categories. The central insight is that internal *stress* of fragile objects, which determines whether an item is damaged but is not measured by any sensor on our robot, can be computed exactly within a soft body simulator. This allows gentleness to be formulated as a planning objective rather than as a signal that must be sensed at run time. A finite-element model predicts the stress induced by a candidate grasp pose and closing width, and a scripted demonstrator selects one that minimizes stress while still holding the item against gravity. We execute this demonstrator across 34 fragile food categories, randomizing object pose, size, shape, and material within each, to collect over 6k demonstrations without human tele-operation. Using the pooled data, we train a single policy and deploy the policy on a 7-DoF arm that observes the scene as a point cloud obtained from a single external depth camera, a sensing modality selected for its narrow sim-to-real appearance gap; no tactile or force sensing is used in the control loop. In real-world experiments, the policy grasps objects from 11 categories with a [XX%] success rate while handling in-category variation.

We make three contributions: (1) a stress-aware grasp planner based on a width-controlled, distributed-pad contact model that selects both the grasp location and closing width for a deformable item; (2) a single policy that handles diverse fragile food categories using only point-cloud observations; and (3) a wide yet rigorous real-world evaluation demonstrating strong performance across multiple categories and in-category variations in pose, size, and shape, over baselines. Together, these contributions represent a step towards broadly applicable gentle manipulation of everyday food.

## II. RELATED WORKS

### A. Gentle manipulation of fragile objects

Damage-free handling of fragile produce is a long-standing goal in agricultural and food automation and remains open across the pre- and post-harvest pipeline [3]. Much of the existing work addresses this problem through end-effector mechanics: compliant, pneumatic, and enveloping grippers limit contact pressure by construction [4], [1], [10]. A recent co-design perspective jointly optimises end-effector morphology and manipulation control under a stress-aware objective, with evaluations on deformable and fragile foods [11]. Online feedback provides a complementary approach: tactile sensors enable fruit recognition and grasp-force prediction for damage-free handling [12], while fracture-anticipation models learn from controlled breakage trials to handle tofu, potato chips, and bananas [13]. These approaches mitigate damage through mechanical design or contact feedback. Our work instead focuses on the source of policy supervision, using a physics-aware objective to select both the grasp pose and closing width for a fixed parallel-jaw gripper.

### B. Grasp quality for deformable objects

Classical analytic grasp metrics assume rigid objects and quantify force closure or robustness to external

wrenches [14], [15]. Such criteria capture whether a grasp can resist disturbances, but not the deformation or damage caused by the grasp itself. For deformable objects, Le *et al.* simulate deformation-dependent contacts to construct a grasp quality metric that predicts grasp success more reliably [16]. Fragile objects impose the additional requirement of limiting internal stress. Pan *et al.* address this requirement with a stress-minimising metric that accounts for object material and identifies fragile regions [17]. However, their formulation assumes point contacts and general multi-contact grasps. For a two-pad parallel-jaw gripper acting on smooth, rounded objects, the resistible-wrench hull becomes degenerate and therefore does not distinguish among our candidate grasps (Section III-B). Finite-element simulation has also been used to characterise deformable grasp outcomes at scale. DefGraspSim evaluates parallel-jaw grasps across a large object set and reports stress, deformation, and stability metrics with good sim-to-real correspondence [18]. Whereas DefGraspSim *evaluates* candidate grasps, we use the same physics to *search* for them and generate the demonstrations on which a policy is trained.

### C. Data diversity in robot learning

Data diversity is a key driver of generalisation in robot learning [19]. In imitation learning, Lin *et al.* find that generalisation follows a power law in the number of environments and objects, whereas the benefit of additional demonstrations per object saturates quickly; coverage therefore matters more than demonstration count alone [7]. The need for coverage becomes even more in high-precision tasks, for which the number of required demonstrations grows super-exponentially as the tolerance tightens [8]. The same principle extends beyond imitation learning [20], [21]. For instance, Play2Perfect pretrains a task-agnostic dexterous policy through reinforcement learning on procedurally generated objects and random goal trajectories [22]. It shows that increasing object diversity and sampling diverse trajectories during pretraining improve transfer to precise assembly. These results demonstrate that diverse experience can be acquired through autonomous interaction, rather than exclusively from human demonstrations.

Generating such diversity on physical systems, however, remains expensive. MimicGen addresses this cost by transforming a few hundred human demonstrations into tens of thousands of episodes with new object placements [9], and related systems extend this approach to bimanual dexterous manipulation [23] and dynamic tasks [24]. Although these methods increase geometric coverage, they retain the grasps selected by a human. Our method instead generates diverse, task-level demonstrations efficiently in simulation. By randomising object pose, size, shape, and material and replanning each grasp with a stress-aware physics objective, the demonstrator adapts both the grasp pose and closing width to every sampled configuration. The resulting supervision therefore captures physical as well as geometric variation without requiring human teleoperation.

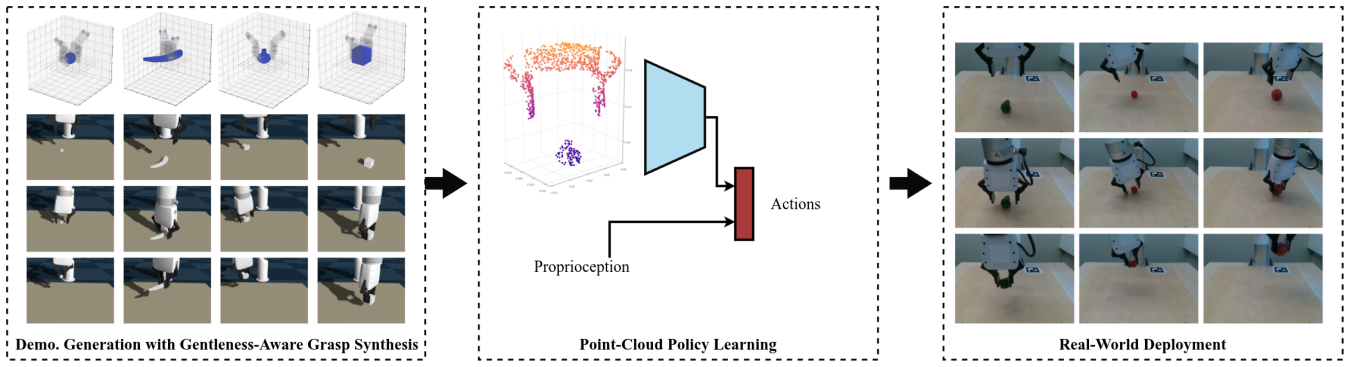


Fig. 2. Method pipeline (Tmp, need improvement)

### III. METHOD

#### A. Problem Setting

We consider a robot arm with a parallel-jaw gripper must lift a single fragile object from a table and hold it without damaging it. Object category, pose, size, shape and material vary across episodes. Perception is provided by a single fixed external depth camera; there is no tactile or force sensing in the loop (Should we justify this? Like simplifying setup and/or smoother sim2real). At each control step the policy observes  $o_t = (\mathbf{p}_t, \mathbf{q}_t, \omega_t, \mathcal{P}_t)$ : end-effector position  $\mathbf{p}_t$ , orientation as a unit quaternion  $\mathbf{q}_t$ , measured gripper width  $\omega_t$ , and a point cloud  $\mathcal{P}_t$  in the robot base frame. It emits an *absolute* target pose and width  $a_t = (\mathbf{p}_t^{\text{cmd}}, \theta_t^{\text{cmd}}, \omega_t^{\text{cmd}}) \in \mathbb{R}^7$ , with orientation as Euler angles. An episode succeeds if the object centre enters a height band. We evaluate gentleness by (1) in simulation, top decile internal von Mises stress stays below the material’s yield threshold (2) in the real world, a damage scale 0-5 evaluated by more than one human evaluator.

#### B. Gentleness-Aware Grasp Synthesis

For each object instance the demonstrator synthesizes a 7-DoF grasp  $x = (\mathbf{t}, \mathbf{r}, \omega)$ : tool-center position  $\mathbf{t}$ , orientation  $\mathbf{r}$ , and *commanded* gripper width  $\omega$ . The grasp pose is executed by a demonstration generator to generate demonstrations in simulation.

The workflow of the synthesizer is three parts. A width-controlled finite-element contact model predicts, for a candidate grasp, the stress it would impose and the force it would hold with; A feasibility ladder then excludes candidates that miss the object, penetrate it, or cannot carry it, and among those that survive an objective trades bulk deformation against local contact pressure. Finally a search — geometric seeding, a batched scoring pass, local optimization, and a width re-scan — returns the gentlest holdable grasp, re-run under progressively looser bounds if the object admits none. We detail each in turn.

**Width-controlled contact model.** Common parallel jaw grippers are position-controlled: one commands a width and the grip force is an outcome. We model this directly. The object is tetrahedralised and a linear-elastic stiffness matrix assembled; since it is free-floating the system is singular in

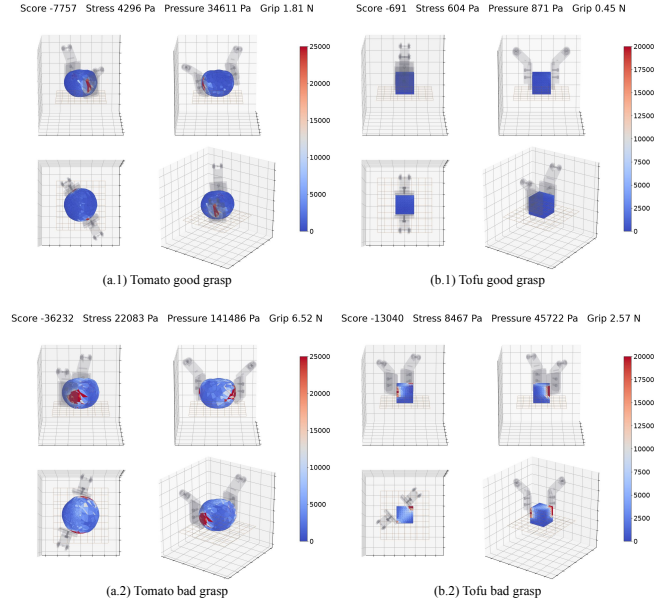


Fig. 3. FEM simulation of different grasps on two objects, tomato and tofu. **TODO: make it clearer.**

the six rigid-body modes, so we solve a bordered inertia-relief system, factored once per geometry. Two rigid pads, whose rectangular footprints are taken from the real gripper’s finger meshes, are closed to width  $\omega$ . Boundary nodes inside a footprint are pushed onto the pad’s flat plane and constrained *only* along the closing axis, leaving tangential motion free so the material bulges; a thin smooth-step fillet at the pad edge avoids a flat-punch stress singularity. Contact is normal-only, and friction enters solely through the holdability test below.

**Material scaling.** Because the displacement is prescribed rather than the force, the deformation field is independent of Young’s modulus, instead is dictated by the grasp width  $\omega$ . Thus below yield both stress and grip force are linear in it,

$$\sigma(E) = E \sigma_1, \quad F(E) = E F_1. \quad (1)$$

One solve at  $E=1$  therefore yields every material by scalar multiplication, which makes randomizing stiffness across the demonstration set cheap. A grasp is *holdable* when Coulomb

TABLE I

FEASIBILITY GATES.  $\delta$  IS THE PAD INDENTATION,  $F$  THE PER-JAW NORMAL FORCE,  $A_{\min}$  THE SMALLER PAD’S CONTACT AREA,  $\hat{\mathbf{a}}$  THE CLOSING AXIS,  $\boldsymbol{\tau}$  THE GRAVITATIONAL TORQUE ABOUT THE OBJECT’S CENTRE OF MASS, AND  $\sigma_Y$  THE MATERIAL’S YIELD STRESS.

| gate        | feasible when                                                                           |
|-------------|-----------------------------------------------------------------------------------------|
| table       | finger meshes stay above the table plane                                                |
| penetration | finger meshes have penetration less than $c$                                            |
| contact     | each pad captures a non-empty contact patch                                             |
| indentation | $\delta \leq \delta_{\max}$ (linear-elastic regime)                                     |
| holdability | $2\mu F \geq m(g + a)$                                                                  |
| torsion     | $ \boldsymbol{\tau} \cdot \hat{\mathbf{a}}  \leq \frac{4}{3} \mu F \sqrt{A_{\min}/\pi}$ |
| yield       | $\sigma_{\text{blk}} \leq \sigma_Y$                                                     |

friction at the two pads carries the object’s weight with an acceleration margin,  $2\mu F(E) \geq m(g + a)$ ; we use  $\mu = 0.7$  and  $a = g$ , a quasi-static  $2g$  criterion rather than a wrench-closure analysis.

**Objective.** A grasp pose candidate is *feasible* when it clears every gate of Table I: the fingers clear the table and the object’s interior, both pads make contact at an indentation the linear-elastic model still supports, and the grasp holds the object against sliding, twisting and yielding. Among feasible candidates, we maximize a score:

$$S(x) = -\sigma_{\text{blk}} - w_{\text{pr}} \frac{F}{A_{\min}}, \quad (2)$$

where  $\sigma_{\text{blk}}$  be the mean of the top decile of element stresses with contact-adjacent elements masked out, so that it reports *bulk* deformation rather than the mesh-dependent contact singularity.  $A_{\min}$  is the contact area of the worse pad, so the second term is the mean contact pressure. Bulk gentleness and local gentleness are thus both represented, and neither has a manually-set floor. Figure 3 shows good and bad valid grasps scored by our FEM model.

**Search.** The search is a funnel (Algorithm 4). Candidates are seeded by two geometric generators — antipodal surface pairs within the friction cone, and medial-axis points closing across the local tangent — each yielding a complete 7-DoF pose with  $\omega$  initialised to the local cross-section. Seeds are filtered on the two cheaply batched gates (table clearance and finger penetration, at a loose tolerance) and on an orientation bound; survivors are then scored and gated (Tab. I) using FEM solvers. The top  $K$  candidates serve as seeds for CMA-ES. The orientation box bounds the search rather than gating it: poses outside are never evaluated, whereas a candidate failing a gate is evaluated and penalised. It holds roll and pitch away from tilts at which the wrist would strike the table, and limits yaw so the fingers do not occlude the object in the view the policy must act on. The CMA-ES searches all seven degrees of freedom jointly. A final pass then fixes each of the best spatially distinct poses and sweeps width alone on a grid finer than the optimizer’s step, retaining the highest-scoring holdable value.

**Relaxation ladder.** No single feasibility band serves an object set spanning three orders of magnitude in stiffness and scale. The funnel is therefore run inside a short ladder:

### Algorithm 1 Stress-aware grasp synthesis

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**Input:** mesh, pose, material  $(E, \rho, \mu)$ , table height  
**Output:** grasp  $x = (\mathbf{t}, \mathbf{r}, \omega)$

- 1 assemble FEM, factor the bordered system *once*
- 2  $X_0 \leftarrow$  antipodal grasps  $\cup$  medial-axis grasps
- 3 **for** tier  $t = 0, 1, 2$  **do**
- 4   bounds  $\leftarrow$  RELAX ( $t$ )
- 5    $X_1 \leftarrow \{x \in X_0 : \text{table, orientation, penetration}\}$
- 6   score  $X_1$ ;  $X_2 \leftarrow$  top  $K$  among  $X_1$
- 7    $X_3 \leftarrow$  CMA-ES ( $X_2$ ) over all 7 DoF
- 8    $X_4 \leftarrow$  spatially distinct poses of  $X_3$
- 9    $x^* \leftarrow \arg \max S(x)$  over a 1-DoF width sweep at each pose of  $X_4$
- 10   **if**  $x^*$  is holdable **then return**  $x^*$
- 11 **return** most-centred feasible grasp seed (a fallback when no holdable solution is found.)

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Fig. 4. The funnel (lines 6–10) is re-run under progressively looser bounds and returns at the first tier that yields a holdable grasp.

the first pass uses nominal bounds; the second widens the orientation box and the penetration tolerance while *doubling* the pressure weight of Eq. (2), so that the constraints loosen and the objective tightens together; the third additionally releases the yield gate, accepting a grasp that marks the object rather than one that fails to lift it. The ladder stops at the first pass yielding a holdable grasp and so costs nothing on objects that do not need it; if none succeeds, the planner returns its most-centred feasible seed and the episode is recorded as an attempt. The final release is a last resort rather than a licence: 2.9% of planned grasps invoked it.

### C. Demonstration Generation

Planned grasps are executed in a deformable (MPM) simulator, producing (observation, action) trajectories in the same format a real teleoperation session yields. The executor runs a phase machine—approach, settle, close, firm, lift, hold.

**Randomization.** Each episode samples object pose, uniform scale, and a procedural shape, together with the material parameters. For each object, we randomize its initial pose, shape, size, and stiffness parameters during collection. For shape randomization, we apply Barr bend [25], twist and taper along the object’s longest axis among the x-y-z axes. Each category additionally carries several meshes representing distinct variants within the type.

**Approach trajectory.** Rather than interpolating linearly from home to the grasp pose, the approach reproduces the structure we measured in human teleoperation: lateral alignment completes while the gripper is still well above the object, and descent continues afterwards. We implement this as per-axis progress within a single phase, with no waypoint and no stop,

$$s_{xy}(\alpha) = \text{smoothstep}(\min(\alpha/f, 1)), \quad s_z(\alpha) = \alpha, \quad (3)$$

with  $0 < \alpha \leq 1$  and  $f$  drawn per environment, setting how early the lateral motion finishes; orientation is slerped over the full  $\alpha$ . Lateral velocity reaches zero only at its own arrival, while  $z$  is still moving, so the path never dwells. The phase duration is set from the profile’s arc length and

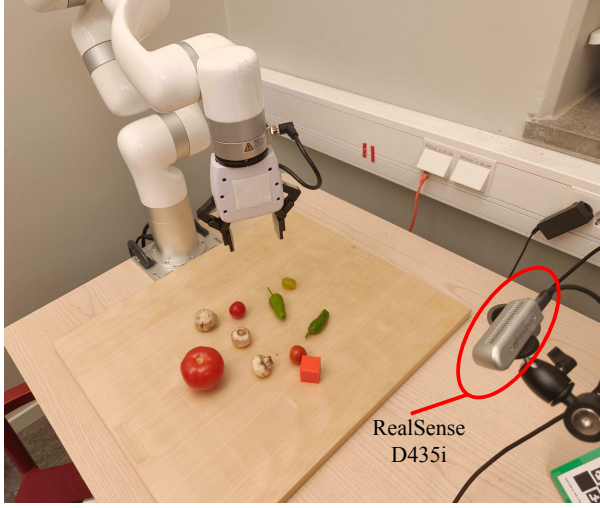


Fig. 5. The real-world experiment workspace setup. A RealSense D435i camera is placed for external view over the workspace. **TODO:** blur background.

a reference speed rather than being fixed, so per-step speed is approximately constant across episodes; a fixed duration would instead make commanded speed depend on how far the object happened to spawn.

#### D. Policy Learning

**TODO:** visualize training object categories (around 30) in sim.

### IV. EXPERIMENT

#### A. Experiment Setup

Experiments use a 7-DoF UFACTORY xArm 7 equipped with its standard two-finger parallel gripper. An external Intel RealSense D435 RGB-D camera positioned in front of the arm observes the workspace.

#### B. Simulation Experiment

**Grasp synthesizer comparison.** Each baseline replaces only the grasp-selection step: the same controller executes the returned pose and width, point-cloud methods see an identical simulated depth observation and mesh-based methods the same tet-mesh surface, and we discard only candidates whose fingers would pass through the table or whose approach exceeds the arm’s orientation limits, executing each method’s own top-ranked survivor unmodified.

- *Naive Top-Down.* Gripper at the object centre, uniformly random yaw, vertical approach, the width is set to the object’s measured cross-section along that axis. It encodes no contact reasoning and establishes the floor of the comparison.
- *Antipodal.* Samples surface point pairs whose connecting line lies inside both friction cones [26] and ranks them by cone margin, commanding the pair’s separation as the width; we rank by margin rather than Ferrari–Canny  $\epsilon$ , which vanishes for two hard point contacts on a curved surface [29].

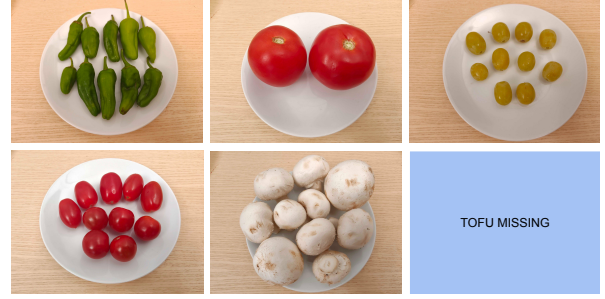


Fig. 6. Real demonstration training set, comprising Padrón peppers, tomatoes, grapes, cherry tomatoes, mushrooms, and tofu, with variation in size and shape within each category. We collect 20 demonstrations per category, summing up to 120 demonstrations for baseline training.

- *GPD* [27]. Samples 6-DoF candidates in the point cloud and scores them with its released CNN. GPD returns, for each candidate, the extent of the observed points enclosed between the fingers—a contact width—which we command directly.
- *GraspNet-1Billion* [28]. Released weights, no algorithmic change; its width is the regressed contact width inflated by a fixed  $1.2\times$  clearance and clipped to the jaw range—a pre-shape opening—which we execute unmodified.
- *Geometry-only (METHOD w/o FEM).* Retains our candidate seeding, CMA-ES search and 1-D width refinement, but substitutes a purely geometric surface-distance objective and omits the FEM-derived feasibility gates. It constitutes an ablation of METHOD without its FEM component.

#### Experiment results.

#### C. Real-World Experiment

**Gentle grasp policy comparison.** We compare against four Diffusion Policy (DP) baselines [30] trained on the real-world demonstrations of object categories in Fig. 6.

- The three RGB baselines use a ResNet-18 encoder: *RGB-frozen* fixes the ImageNet-pretrained encoder and trains only the policy head; *RGB-finetune* jointly trains the pretrained encoder and policy head; and *RGB-scratch* trains both from random initialization.
- *PCD* follows 3D Diffusion Policy [31], using point-cloud observations and a PointNet encoder trained on the real demonstrations.

**TODO:** indicate which categories are unseen and which are in training set. The evaluation set includes six categories represented in the real demonstration set (pepper, tomato, grape, cherry tomato, mushroom, and tofu) and five absent from it (banana, plum, bread, beverage can, and paper cup). **TODO:** Confirm which evaluation categories are also absent from simulation training before claiming generalization to unseen categories.

**Evaluation protocol.** Two evaluators, blinded to the policy identity, assess each rollout under similar lighting conditions. **TODO:** cite. Both receive the same category-specific scoring

TABLE II  
GRASP SYNTHESIS COMPARISON ON 5 DEFORMABLE OBJECTS IN MPM SIMULATION.

| Method                         | Tofu         |            |                         | Mushroom     |            |                         | Strawberry  |            |                         | Pepper      |           |                         | Cherry tomato |           |                         | Mean Succ   | Mean Sub-y  |
|--------------------------------|--------------|------------|-------------------------|--------------|------------|-------------------------|-------------|------------|-------------------------|-------------|-----------|-------------------------|---------------|-----------|-------------------------|-------------|-------------|
|                                | Succ         | Sub-y      | $\bar{\sigma}/\sigma_y$ | Succ         | Sub-y      | $\bar{\sigma}/\sigma_y$ | Succ        | Sub-y      | $\bar{\sigma}/\sigma_y$ | Succ        | Sub-y     | $\bar{\sigma}/\sigma_y$ | Succ          | Sub-y     | $\bar{\sigma}/\sigma_y$ |             |             |
| Naive top-down                 | 13.3         | 100        | 0.23                    | 71.7         | 100        | 0.20                    | 25.0        | 100        | 0.27                    | 5.0         | 100       | 0.23                    | 86.7          | 100       | 0.62                    | 40.3        | 100.0       |
| Antipodal [26]                 | 88.3         | 100        | 0.28                    | 90.0         | 30         | 1.28                    | 76.7        | 100        | 0.38                    | 83.3        | 75        | 0.84                    | 100.0         | 73        | 0.79                    | 87.7        | 75.5        |
| GPD [27]                       | —            | —          | —                       | —            | —          | —                       | —           | —          | —                       | —           | —         | —                       | —             | —         | —                       | —           | —           |
| GraspNet [28]                  | —            | —          | —                       | —            | —          | —                       | —           | —          | —                       | —           | —         | —                       | —             | —         | —                       | —           | —           |
| Geometry-only (METHOD w/o FEM) | 85.0         | 100        | 0.27                    | 100.0        | 30         | 1.08                    | 100.0       | 45         | 1.03                    | 98.3        | 19        | 1.11                    | 81.7          | 2         | 1.20                    | <b>93.0</b> | 39.2        |
| METHOD                         | <b>100.0</b> | <b>100</b> | <b>0.29</b>             | <b>100.0</b> | <b>100</b> | <b>0.32</b>             | <b>96.7</b> | <b>100</b> | <b>0.41</b>             | <b>73.3</b> | <b>98</b> | <b>0.67</b>             | <b>83.3</b>   | <b>51</b> | <b>0.98</b>             | 90.7        | <b>89.7</b> |

rubrics before evaluation and record quantitative scores and qualitative observations.

Each evaluator assigns two scores in increments of 0.5. The lift score  $s_{\text{lift}} \in [0, 5]$  is zero if the object is not lifted and five if it reaches at least 20 cm above the tabletop; intermediate scores reflect the estimated lift height. The damage score  $s_{\text{damage}} \in [0, 5]$  assesses damage throughout approach, grasp, lift, and hold: zero denotes an intact object, and five denotes severe damage, such as crushing or irreversible deformation. Intermediate damage scores follow empirically defined category-specific rubrics. **TODO: Add the damage-rubric example figure and specify the height-to-score mapping and hold duration.**

We derive three normalized metrics from each evaluator’s scores:

$$\begin{aligned} \text{Lift} &= s_{\text{lift}}/5, \\ r &= \begin{cases} -s_{\text{damage}}, & s_{\text{lift}} = 0, \\ 5 - s_{\text{damage}}, & s_{\text{lift}} > 0, \end{cases} \\ \text{Safe} &= 0.5 + 0.1r, \\ \text{S-Lift} &= (\text{Lift} + \text{Safe})/2. \end{aligned}$$

All three lie in  $[0, 1]$ , with higher values indicating better performance. Safe penalizes damage and assigns an intact, unlifted object a score of 0.5; it is a lift-conditioned score, not an undamaged-trial rate. S-Lift weights lifting and safety equally. We average the derived metrics over evaluators and trials within each category, then equally over categories.

Trials use a  $26 \times 30$  cm workspace, with 55 trials per method distributed across 11 categories as listed in Table III. Each method uses a separate set of object instances to avoid effects of prior manipulation; instances vary in size, shape, and fragility within each category. Initial poses are randomized within the workspace, with the same pose used across methods for each corresponding trial index within a category.

**Experiment results.** Table III reports the results on 11 object categories, averaged over two evaluators. The Mean column in the first block reports the three scores averaged across all 11 categories. Our method achieves the highest category-averaged Lift, Safe, and S-Lift scores (0.66, 0.78, and 0.72), compared with the best baseline scores of 0.57, 0.67, and 0.62, respectively. The gains are not uniform across categories: baselines obtain higher S-Lift scores on banana, tofu, bread, and grape.



Fig. 7. Evaluation sets covering 11 categories: tofu, banana, pepper, cherry tomato, mushroom, bread, tomato, beverage can, paper cup, grape, and plum. Each method uses the same number of instances per category, with separate instances across methods. **TODO: color/contrast rebalance.**

## V. CONCLUSIONS

This work presents a stress-aware framework for learning gentle visuomotor grasping policies. Its core component is a grasp synthesis algorithm that uses simulated internal stress to generate demonstrations for fragile objects at scale. Policies trained on these demonstrations transfer zero-shot to the real world without tactile sensing or real-world policy demonstrations. Simulation experiments demonstrate improved grasp synthesis performance, while real-world evaluation yields higher lifting and safety scores than policies trained on real demonstrations.

The demonstration dataset remains limited in category coverage; expanding it requires balancing data across categories. Future work will investigate simulation–real co-training to improve deployment robustness. We also introduce a multi-evaluator protocol with category-specific rubrics for assessing object damage alongside lifting performance. Automated evaluators could reduce assessment effort, although partial visual observations may limit their ability to detect damage.

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TABLE III  
REAL-WORLD MANIPULATION RESULTS. EACH RESULT CELL REPORTS LIFT|SAFE|S-LIFT.

| Method<br># Trials  | Mean<br>55 |      |             | Banana<br>4 |      |             | Tofu<br>7 |      |             | Pepper<br>2 |      |             | Plum<br>2 |      |             | Tomato<br>2 |      |             |
|---------------------|------------|------|-------------|-------------|------|-------------|-----------|------|-------------|-------------|------|-------------|-----------|------|-------------|-------------|------|-------------|
| <i>RGB-frozen</i>   | 0.18       | 0.53 | 0.36        | 0.00        | 0.50 | 0.25        | 0.00      | 0.50 | 0.25        | 0.00        | 0.50 | 0.25        | 0.50      | 0.65 | 0.58        | 0.00        | 0.50 | 0.25        |
| <i>RGB-finetune</i> | 0.46       | 0.67 | 0.56        | 0.00        | 0.50 | 0.25        | 0.00      | 0.50 | 0.25        | 1.00        | 0.95 | <b>0.98</b> | 0.75      | 0.80 | 0.78        | 1.00        | 0.96 | <b>0.98</b> |
| <i>RGB-scratch</i>  | 0.57       | 0.67 | 0.62        | 0.50        | 0.61 | 0.56        | 0.18      | 0.61 | 0.39        | 1.00        | 0.95 | <b>0.98</b> | 0.50      | 0.71 | 0.61        | 0.50        | 0.68 | 0.59        |
| <i>PCD</i>          | 0.43       | 0.67 | 0.55        | 0.50        | 0.63 | <b>0.56</b> | 0.29      | 0.63 | <b>0.46</b> | 0.50        | 0.75 | 0.63        | 0.50      | 0.71 | 0.61        | 0.50        | 0.70 | 0.60        |
| <i>Ours</i>         | 0.66       | 0.78 | <b>0.72</b> | 0.25        | 0.60 | 0.43        | 0.19      | 0.60 | 0.39        | 0.95        | 1.00 | <b>0.98</b> | 1.00      | 0.90 | <b>0.95</b> | 1.00        | 0.93 | 0.96        |

| Method<br># Trials  | Mushroom<br>6 |      |             | Bread<br>4 |      |             | Cherry tomato<br>10 |      |             | Grape<br>5 |      |             | Can<br>5 |      |             | Paper cup<br>8 |      |             |
|---------------------|---------------|------|-------------|------------|------|-------------|---------------------|------|-------------|------------|------|-------------|----------|------|-------------|----------------|------|-------------|
| <i>RGB-frozen</i>   | 0.33          | 0.53 | 0.43        | 0.00       | 0.50 | 0.25        | 0.01                | 0.52 | 0.26        | 0.02       | 0.45 | 0.24        | 0.60     | 0.58 | 0.59        | 0.50           | 0.62 | 0.56        |
| <i>RGB-finetune</i> | 0.33          | 0.62 | 0.48        | 0.25       | 0.59 | 0.42        | 0.30                | 0.63 | 0.47        | 0.00       | 0.50 | 0.25        | 0.80     | 0.68 | 0.74        | 0.60           | 0.65 | 0.63        |
| <i>RGB-scratch</i>  | 0.83          | 0.65 | 0.74        | 0.25       | 0.61 | 0.43        | 0.55                | 0.72 | 0.64        | 0.60       | 0.71 | <b>0.65</b> | 0.78     | 0.58 | 0.68        | 0.58           | 0.54 | 0.56        |
| <i>PCD</i>          | 0.67          | 0.75 | 0.71        | 0.53       | 0.73 | <b>0.63</b> | 0.20                | 0.58 | 0.39        | 0.40       | 0.70 | 0.55        | 0.00     | 0.50 | 0.25        | 0.63           | 0.64 | 0.63        |
| <i>Ours</i>         | 0.83          | 0.86 | <b>0.85</b> | 0.25       | 0.63 | 0.44        | 0.60                | 0.77 | <b>0.69</b> | 0.20       | 0.60 | 0.40        | 1.00     | 0.88 | <b>0.94</b> | 1.00           | 0.86 | <b>0.93</b> |

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